Multilevel Models (part 2)

EES 4891-06/5891-01

Bayesian Statistical Methods

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Class #19: Wednesday March 25, 2026

Chimpanzees

Chimpanzee Experiment

- Experimental treatments:
  - Partner or not
  - Pro-social lever on left or right
- Other conditions
  - Experiments were performed in “blocks” during different days
- Outcome:
  - L: Which lever did the chimp pull (1 = left, 0 = right)

- Load data

```
library(rethinking)
data(chimpanzees)
d <- chimpanzees |>
  mutate(treatment = 1L + prosoc_left + 2L * condition)

dat_list <- select(d, pulled_left, actor, block_id = block,
                   treatment)
```

Original Model

- Original Model:

$$
\begin{aligned}
L &\sim \text{Binomial}(1, p) \\
\text{logit}(p) &= \alpha_{\text{actor}} + \beta_{\text{treatment}} \\
\alpha &\sim dnorm(0, 1.5) \\
\beta &\sim dnorm(0, 0.5)
\end{aligned}
$$

  - Limitations:
    - Prior for α assumes equal likelihood of being left- or right-handed
    - Experiments were conducted in 6 “blocks” on different days. Did chimpanzees behave differently on different days?

- Model code:

```
set.seed(25)
mdl_chimp.1 <- ulam(
    alist(
        pulled_left ~ dbinom(1, p),
        logit(p) <- a[actor] + b[treatment],
        b[treatment] ~ dnorm(0, 0.5),
        a[actor] ~ dnorm(0, 1.5)
    ), data=dat_list, chains = 4, cores = 4, log_lik = TRUE)
```

Multilevel Model

- New model:

$$
\begin{aligned}
L &\sim \text{Binomial}(1, p) \\
\text{logit}(p) &= \alpha_{\text{actor}} + \gamma_{\text{block}} + \beta_{\text{treatment}} \\
\beta_{\text{treatment}} &\sim \text{Normal}(0, 0.5), \text{treatment in } 1 \ldots 4 \\
\alpha_{\text{actor}} &\sim \text{Normal}(\bar{\alpha}, \sigma_\alpha), \text{actor in } 1 \ldots 7 \\
\gamma_{\text{block}} &\sim \text{Normal}(0, \sigma_\gamma), \text{block in } 1 \ldots 6 \\
\bar{\alpha} &\sim \text{Normal}(0, 1.5) \\
\sigma_\alpha &\sim \text{Exponential}(1) \\
\sigma_\gamma &\sim \text{Exponential}(1)
\end{aligned}
$$

- This allows for partial pooling across actors and blocks (e.g., most actors tend to pull the right lever, but some prefer the left)
- Example of a multilevel model with two different kinds of clusters.

- Model code:

```
set.seed(13)
mdl_chimp.2 <- ulam(
    alist(
        pulled_left ~ dbinom(1, p),
        logit(p) <- a[actor] + g[block_id] + b[treatment],
        b[treatment] ~ dnorm(0, 0.5),
      ## adaptive priors
        a[actor] ~ dnorm(a_bar, sigma_a),
        g[block_id] ~ dnorm(0, sigma_g),
      ## hyper-priors
        a_bar ~ dnorm(0, 1.5),
        sigma_a ~ dexp(1),
        sigma_g ~ dexp(1)
    ), data=dat_list, chains = 4, cores = 4, log_lik = TRUE)
```

- Hyperparameters: $\bar{\alpha}, \sigma_\alpha, \sigma_\gamma$
- Hyperpriors for the hyperparameters:

$$
\begin{aligned}
\bar{\alpha} &\sim \text{Normal}(0, 1.5) \\
\sigma_\alpha &\sim \text{Exponential}(1) \\
\sigma_\gamma &\sim \text{Exponential}(1)
\end{aligned}
$$

Why We Must Use MCMC Methods

- Quadratic approximation (`quap`) works well for completely pooled and completely un-pooled data
  - Find maximum of probability density by varying parameters one at a time.
  - Multilevel model: The effect of varying one parameter depends on the values of other hyperparameters.
    - Varying parameters and hyperparameters one at a time won't efficiently find the maximum.
- Hamiltonian Monte Carlo (`ulam`): imaginary particle moving in N-dimensional space, varying all parameters simultaneously.

Model Results

- Model 1 (the old one)

```
precis(mdl_chimp.1, depth = 2, digits = 2)
```

```
##       mean   sd  5.5% 94.5% rhat ess_bulk
## b[1] -0.01 0.28 -0.48  0.42    1   617.56
## b[2]  0.52 0.29  0.05  0.99    1   670.66
## b[3] -0.35 0.28 -0.79  0.08    1   717.91
## b[4]  0.40 0.28 -0.06  0.85    1   697.56
## a[1] -0.48 0.34 -0.99  0.07    1   800.53
## a[2]  3.88 0.77  2.72  5.24    1  1160.62
## a[3] -0.79 0.33 -1.31 -0.26    1   786.29
## a[4] -0.78 0.33 -1.31 -0.26    1   785.30
## a[5] -0.49 0.32 -1.00  0.04    1   712.42
## a[6]  0.44 0.34 -0.07  0.99    1   790.12
## a[7]  1.93 0.42  1.30  2.64    1  1044.93
```

- Model 2 (multilevel)

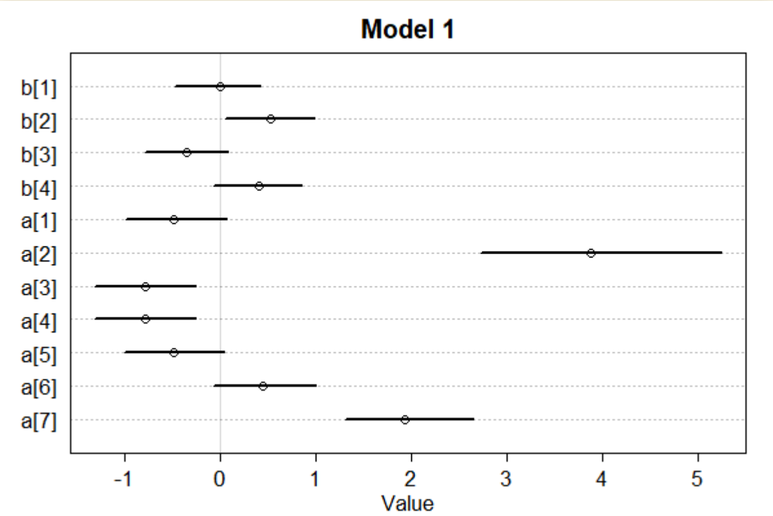
```
precis(mdl_chimp.2, depth = 2, digits = 2)
```

```
##          mean   sd  5.5% 94.5% rhat ess_bulk
## b[1]    -0.12 0.28 -0.58  0.31 1.01   616.79
## b[2]     0.40 0.29 -0.05  0.87 1.01   594.26
## b[3]    -0.47 0.29 -0.96 -0.02 1.00   563.73
## b[4]     0.29 0.29 -0.17  0.75 1.01   488.59
## a[1]    -0.35 0.34 -0.90  0.19 1.01   502.88
## a[2]     4.73 1.38  3.05  7.31 1.01   467.66
## a[3]    -0.66 0.36 -1.24 -0.08 1.01   601.74
## a[4]    -0.68 0.35 -1.23 -0.10 1.00   493.56
## a[5]    -0.36 0.34 -0.91  0.19 1.01   463.32
## a[6]     0.58 0.35  0.03  1.13 1.02   552.05
## a[7]     2.13 0.47  1.40  2.87 1.01   536.20
## g[1]    -0.17 0.23 -0.59  0.08 1.02   306.83
## g[2]     0.03 0.18 -0.25  0.35 1.02   869.80
## g[3]     0.04 0.19 -0.22  0.36 1.02  1146.86
## g[4]     0.00 0.19 -0.30  0.31 1.03  1089.58
## g[5]    -0.03 0.18 -0.35  0.23 1.01   930.26
## g[6]     0.10 0.19 -0.14  0.44 1.02   482.86
## a_bar    0.62 0.72 -0.53  1.84 1.01   926.48
## sigma_a  2.02 0.64  1.21  3.17 1.01   665.96
## sigma_g  0.21 0.18  0.02  0.53 1.05    82.00
```

Plotting Model Results

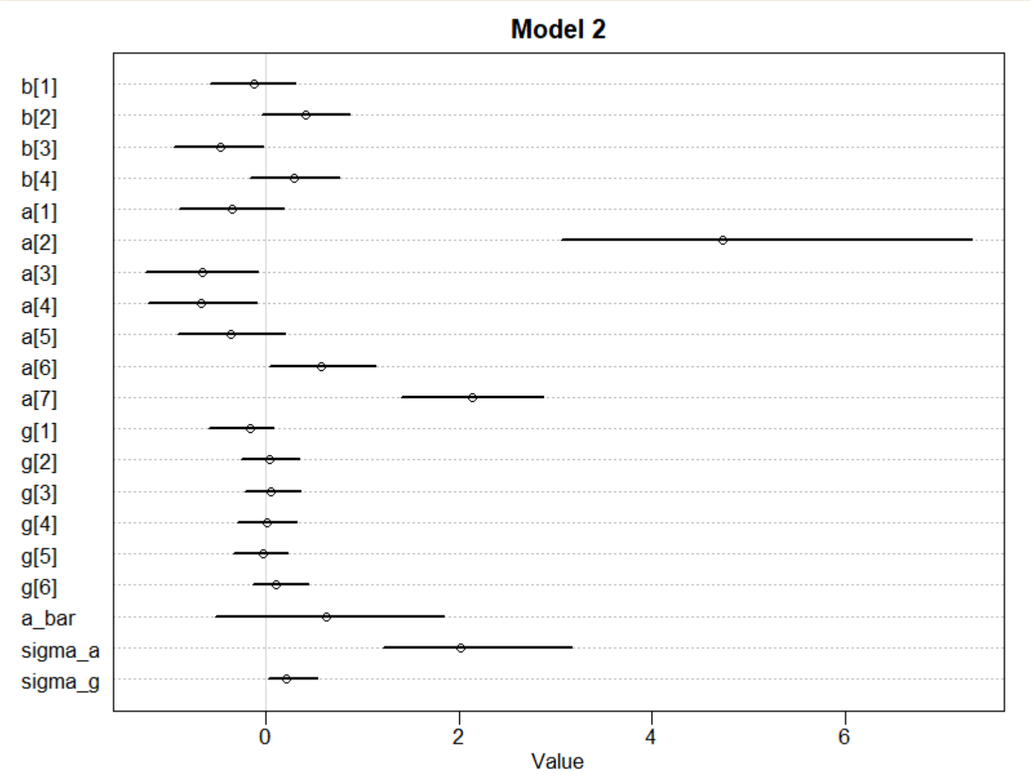
- Model 1

```
plot(precis(mdl_chimp.1, depth = 2, digits = 2), main = "Model
        1")
```



- Model 2

```
plot(precis(mdl_chimp.2, depth = 2, digits = 2), main = "Model
        2")
```



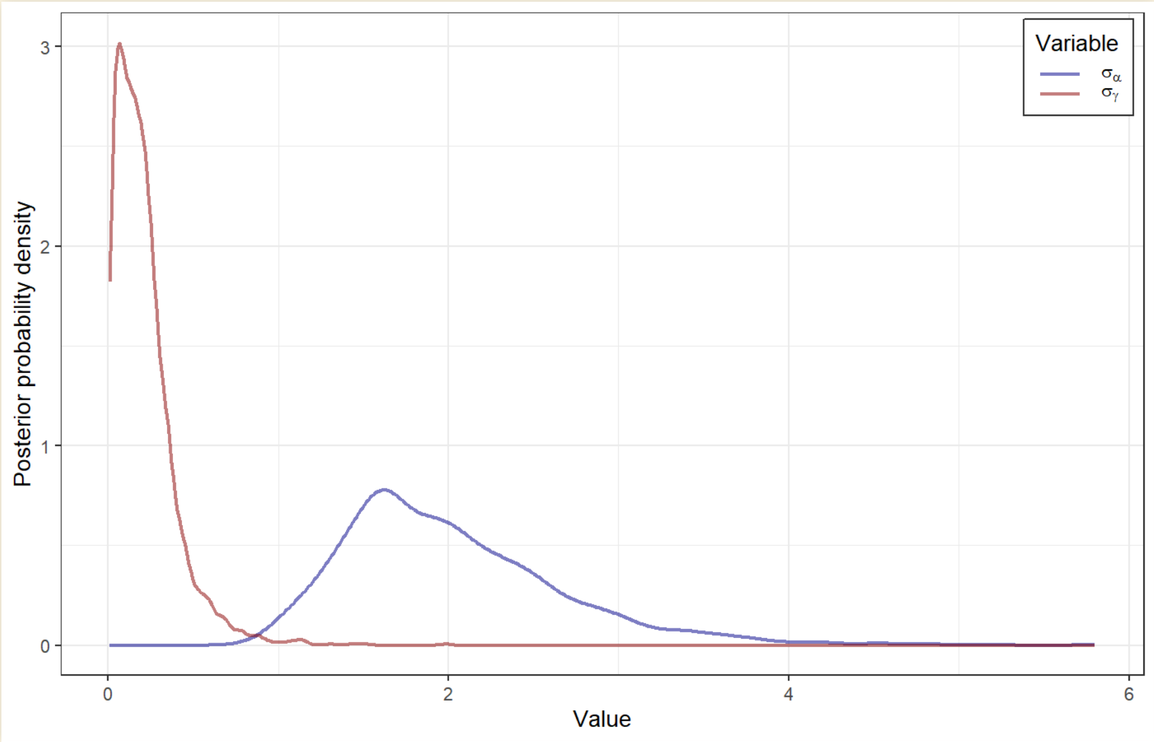
  - All the γ coefficients are consistent with 0.

Actors vs. Blocks

- Model:

$$
\begin{aligned}
L &\sim \text{Binomial}(1, p) \\
\text{logit}(p) &= \alpha_{\text{actor}} + \gamma_{\text{block}} + \beta_{\text{treatment}} \\
\beta_{\text{treatment}} &\sim \text{Normal}(0, 0.5), \text{ treatment in } 1 \ldots 4 \\
\alpha_{\text{actor}} &\sim \text{Normal}(\bar{\alpha}, \sigma_\alpha), \text{ actor in } 1 \ldots 7 \\
\gamma_{\text{block}} &\sim \text{Normal}(0, \sigma_\gamma), \text{ block in } 1 \ldots 6
\end{aligned}
$$

  - σ_α: variation among actors
  - σ_γ: variation among blocks.
  - Plot the posterior probability distribution for σ_α and σ_γ



- σ_α is clearly greater than zero
- σ_γ is piled up near zero.
  - All the values of γ (block-level coefficients) are consistent with zero.
  - Do blocks even matter?

Another model

- Try a simpler model:
  - No block-level effects

$$
\begin{aligned}
L &\sim \text{Binomial}(1, p) \\
\text{logit}(p) &= \alpha_{\text{actor}} + \beta_{\text{treatment}} \\
\beta_{\text{treatment}} &\sim \text{Normal}(0, 0.5), \text{treatment in } 1 \ldots 4 \\
\alpha_{\text{actor}} &\sim \text{Normal}(\bar{\alpha}, \sigma_\alpha), \text{actor in } 1 \ldots 7 \\
\bar{\alpha} &\sim \text{Normal}(0, 1.5) \\
\sigma_\alpha &\sim \text{Exponential}(1)
\end{aligned}
$$

- Model code:

```
set.seed(13)
mdl_chimp.3 <- ulam(
    alist(
        pulled_left ~ dbinom(1, p),
        logit(p) <- a[actor] + b[treatment],
        b[treatment] ~ dnorm(0, 0.5),
      ## adaptive priors
        a[actor] ~ dnorm(a_bar, sigma_a),
      ## hyper-priors
        a_bar ~ dnorm(0, 1.5),
        sigma_a ~ dexp(1)
    ), data=dat_list, chains = 4, cores = 4, log_lik = TRUE)
```

Compare the models

- Compare the models:

```
compare(mdl_chimp.1, mdl_chimp.2, mdl_chimp.3)
```

```
##               WAIC    SE dWAIC  dSE pWAIC weight
## mdl_chimp.3 530.97 19.22  0.00   NA  8.47   0.47
## mdl_chimp.1 532.05 18.99  1.08 0.44  8.43   0.27
## mdl_chimp.2 532.20 19.44  1.22 1.60 10.59   0.26
```

- Model 3 works is slightly better than the original model, and model 2 (with the block clusters) is worst.
  - What is the difference between models 1 and 3?
    - Model 3 has an adaptive prior for α (the actor's preference for right or left).
  - What is the difference between models 2 and 3?
    - Model 3 has one kind of cluster (actor).
    - Model 2 adds a second kind of cluster (block ID)
    - Model 2 overfits the data, because block ID doesn't add any useful information for predictions. (see the pWAIC penalty).

Divergent Transitions

Divergent Transitions

- Sometimes you get warnings about “divergent transitions” when you fit a model using Stan.
- Hamiltonian Monte Carlo (HMC) works by simulating a particle flowing over a surface with hills and valleys.
- In a real physical system, energy is conserved
  - In a computational simulation, steep slopes can cause errors
- When there is a divergent transition, Stan discards it and tries again
  - If there is a region in the posterior where there are a lot of divergent transitions, Stan can’t sample it well.
  - If there is a lot of probability density there, it means you may miss an important part of the posterior.

Dealing with Divergences

- There are two ways to deal with divergences:
  1. Make `adapt_delta` closer to 1.
     - `adapt_delta` is the target acceptance rate used in the tuning (warm-up) phase to adjust the parameters.
     - Default is 0.95. 1 is the maximum allowed value.
     - Setting `adapt_delta` closer to 1 leads to more accurate sampling, but slows Stan down.
  2. Re-write your model to avoid divergences (e.g., re-parameterize it)
     - A common strategy is to change from centered to non-centered parameterizations.
       - **Centered:** Priors for parameters depend directly on other hyperparameters.
       - **Non-centered:** Draw parameters from priors with fixed hyperparameters,
         - Generate *transformed parameters* by scaling the raw hyperparameters
     - Rule of thumb:
       - Non-centered parameterizations are better when there is little data.
       - Centered parameterizations are better when there is a lot of data.

The Devil's Funnel

- Consider this model:

$$\nu \sim \text{Normal}(0, 3)$$
$$x \sim \text{Normal}(0, \exp(\nu))$$

```
mdl_funnel <- ulam(
    alist(
        v ~ normal(0, 3),
        x ~ normal(0, exp(v))
    ), data = list(N = 1), chains = 4, cores = 4)
```

```
precis(mdl_funnel, digits = 2)
```

```
##     mean     sd    5.5% 94.5% rhat ess_bulk
## v   1.93   2.23   -0.85  6.34 1.21    14.51
## x -70.15 395.41 -252.73 24.22 1.21    98.44
```

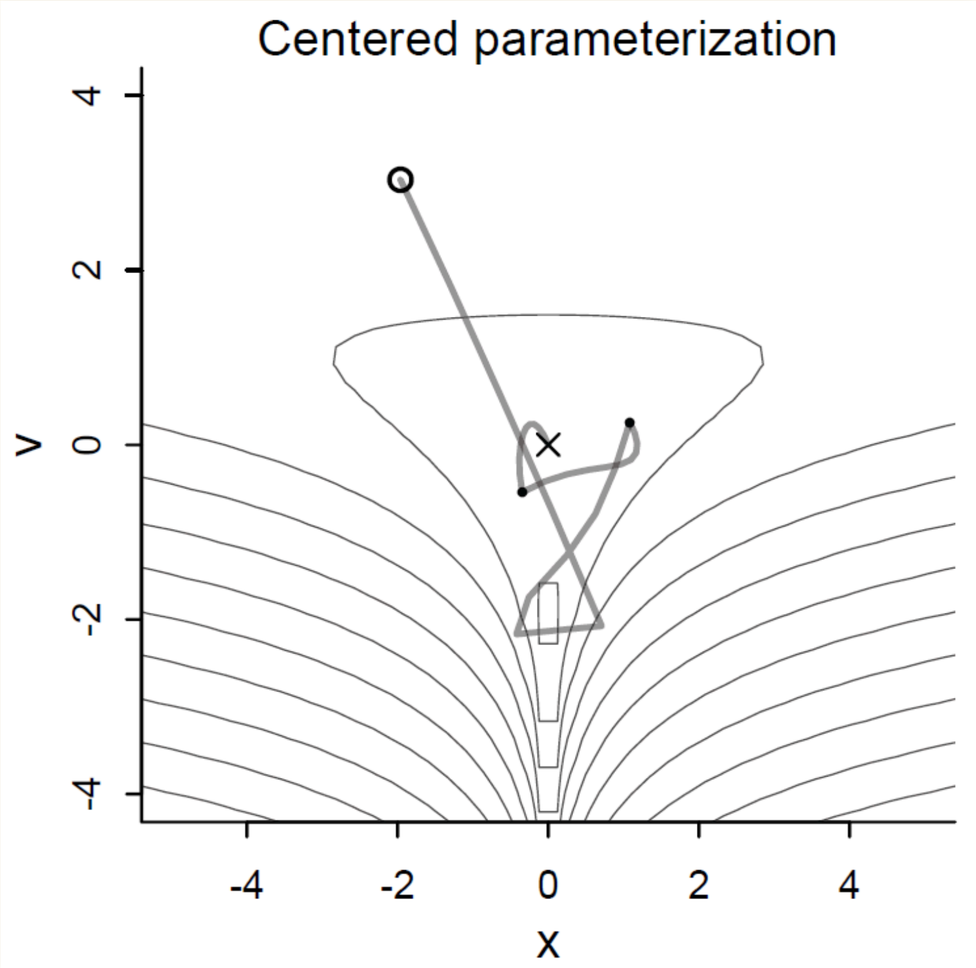
```
mdl_funnel@cstanfit$diagnostic_summary("divergences")
```

```
## Warning: 87 of 2000 (4.0%) transitions ended with a
divergence.
## See https://mc-stan.org/misc/warnings for details.
```

```
## $num_divergent
## [1] 15 40 26  6
```

Cause of Divergences

- Why do divergences happen?
- Potential-energy surface:



- Trajectory starts at “X”, and bounces around, …
  - but when it gets to the steep part near (0, -2) a divergent transition makes it jump way out to the “O”.

Non-Centered Parameterization

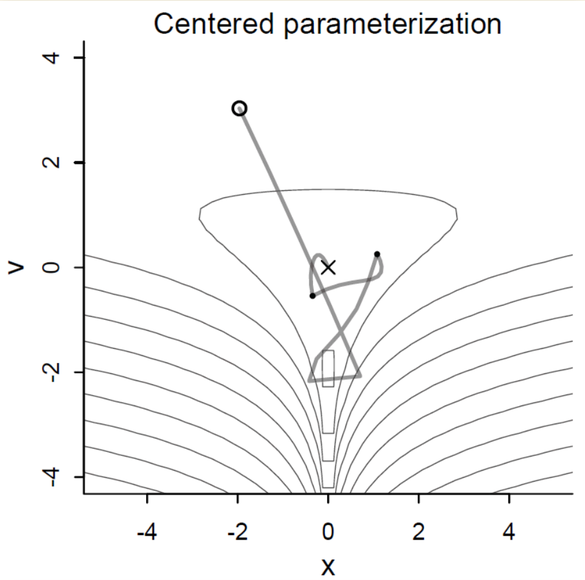
- Original (centered) model:

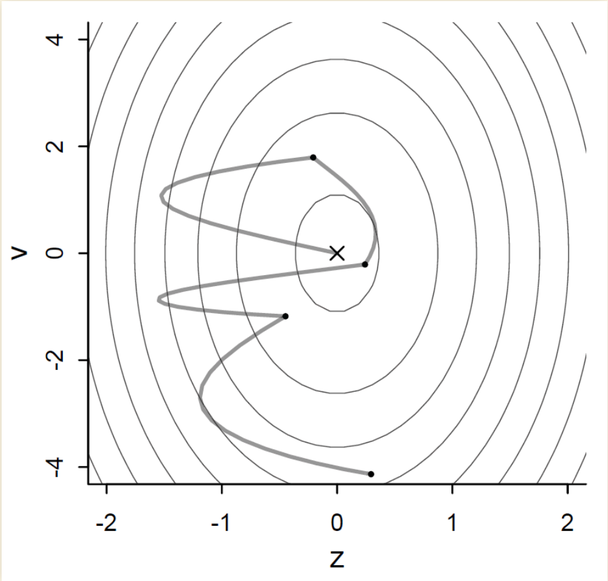
$$\nu \sim \text{Normal}(0, 3)$$
$$x \sim \text{Normal}(0, \exp(\nu))$$

  Prior for x depends on ν
- Non-centered version:

$$\nu \sim \text{Normal}(0, 3)$$
$$z \sim \text{Normal}(0, 1)$$
$$x = z \exp(\nu)$$

  Prior for z is independent of ν, generate x by scaling z.
- This flattens the steep parts by sampling from two Normal distributions and the steep stuff only shows up when we multiply them together.





Coding the model

- Original model:

```
mdl_funnel <- ulam(
    alist(
        v ~ normal(0, 3),
        x ~ normal(0, exp(v))
    ), data = list(N = 1), chains = 4, cores = 4)

precis(mdl_funnel, digits = 2)
```

```
##     mean     sd    5.5% 94.5% rhat ess_bulk
## v   1.93   2.23   -0.85  6.34 1.21    14.51
## x -70.15 395.41 -252.73 24.22 1.21    98.44
```

```
mdl_funnel@cstanfit$diagnostic_summary("divergences")
```

```
## Warning: 87 of 2000 (4.0%) transitions ended with a
divergence.
## See https://mc-stan.org/misc/warnings for details.
```

```
## $num_divergent
## [1] 15 40 26  6
```

- Non-centered model:

```
mdl_funnel_nc <- ulam(
    alist(
        v ~ normal(0, 3),
        z ~ normal(0, 1),
        gq> real[1]:x <<- z*exp(v)
    ), data=list(N=1) , chains=4 )

precis(mdl_funnel_nc, depth = 2, digits = 2)
```

```
##     mean      sd   5.5% 94.5% rhat ess_bulk
## v   0.03    3.12  -4.90  4.85 1.00  1503.28
## z   0.00    0.98  -1.58  1.50 1.00  1218.99
## x -38.58 2237.40 -28.03 24.71 1.01  1393.42
```

```
mdl_funnel_nc@cstanfit$diagnostic_summary("divergences")
```

```
## $num_divergent
## [1] 0 0 0 0
```

Non-Centered Chimpanzees

Divergent Chimpanzee Model

```
set.seed(13)
mdl_chimp.2 <- ulam(
    alist(
        pulled_left ~ dbinom(1, p),
        logit(p) <- a[actor] + g[block_id] + b[treatment],
        b[treatment] ~ dnorm(0, 0.5),
      ## adaptive priors
        a[actor] ~ dnorm(a_bar, sigma_a),
        g[block_id] ~ dnorm(0, sigma_g),
      ## hyper-priors
        a_bar ~ dnorm(0, 1.5),
        sigma_a ~ dexp(1),
        sigma_g ~ dexp(1)
    ), data=dat_list, chains = 4, cores = 4, log_lik = TRUE)

precis(mdl_chimp.2, digits = 2)
```

```
##         mean   sd  5.5% 94.5% rhat ess_bulk
## a_bar   0.62 0.72 -0.53  1.84 1.01   926.48
## sigma_a 2.02 0.64  1.21  3.17 1.01   665.96
## sigma_g 0.21 0.18  0.02  0.53 1.05    82.00
```

```
mdl_chimp.2@cstanfit$diagnostic_summary("divergences")
```

```
## Warning: 11 of 2000 (1.0%) transitions ended with
a divergence.
## See https://mc-stan.org/misc/warnings for
details.
```

```
## $num_divergent
## [1]  0  0 11  0
```

- Increase `adapt_delta` from 0.95 to 0.99:

```
set.seed(13)
mdl_chimp.4 <- ulam(mdl_chimp.2, chains = 4, cores = 4,
                    control = list(adapt_delta = 0.99))

precis(mdl_chimp.4, digits = 2)
```

```
##         mean   sd  5.5% 94.5% rhat ess_bulk
## a_bar   0.56 0.74 -0.59  1.76 1.00   896.99
## sigma_a 2.03 0.69  1.22  3.26 1.00   851.11
## sigma_g 0.18 0.18  0.01  0.50 1.03   107.82
```

```
mdl_chimp.4@cstanfit$diagnostic_summary("divergences")
```

```
## Warning: 9 of 2000 (0.0%) transitions ended with
a divergence.
## See https://mc-stan.org/misc/warnings for
details.
```

```
## $num_divergent
## [1] 0 4 5 0
```

Noncentered Chimpanzee Model

- Original model:

$$
\begin{aligned}
L &\sim \text{Binomial}(1, p) \\
\text{logit}(p) &= \alpha_{\text{actor}} + \gamma_{\text{block}} + \beta_{\text{treatment}} \\
\beta_{\text{treatment}} &\sim \text{Normal}(0, 0.5) \\
\alpha_{\text{actor}} &\sim \text{Normal}(\bar{\alpha}, \sigma_\alpha) \\
\gamma_{\text{block}} &\sim \text{Normal}(0, \sigma_\gamma) \\
\bar{\alpha} &\sim \text{Normal}(0, 1.5) \\
\sigma_\alpha &\sim \text{Exponential}(1) \\
\sigma_\gamma &\sim \text{Exponential}(1)
\end{aligned}
$$

- Non-centered model:

$$
\begin{aligned}
L &\sim \text{Binomial}(1, p) \\
\text{logit}(p) &= \bar{\alpha} + z_{\text{actor}}\sigma_\alpha + x_{\text{block}}\sigma_\gamma + \beta_{\text{treatment}} \\
\beta_{\text{treatment}} &\sim \text{Normal}(0, 0.5) \\
z_{\text{actor}} &\sim \text{Normal}(0, 1) \\
x_{\text{block}} &\sim \text{Normal}(0, 1) \\
\bar{\alpha} &\sim \text{Normal}(0, 1.5) \\
\sigma_\alpha &\sim \text{Exponential}(1) \\
\sigma_\gamma &\sim \text{Exponential}(1)
\end{aligned}
$$

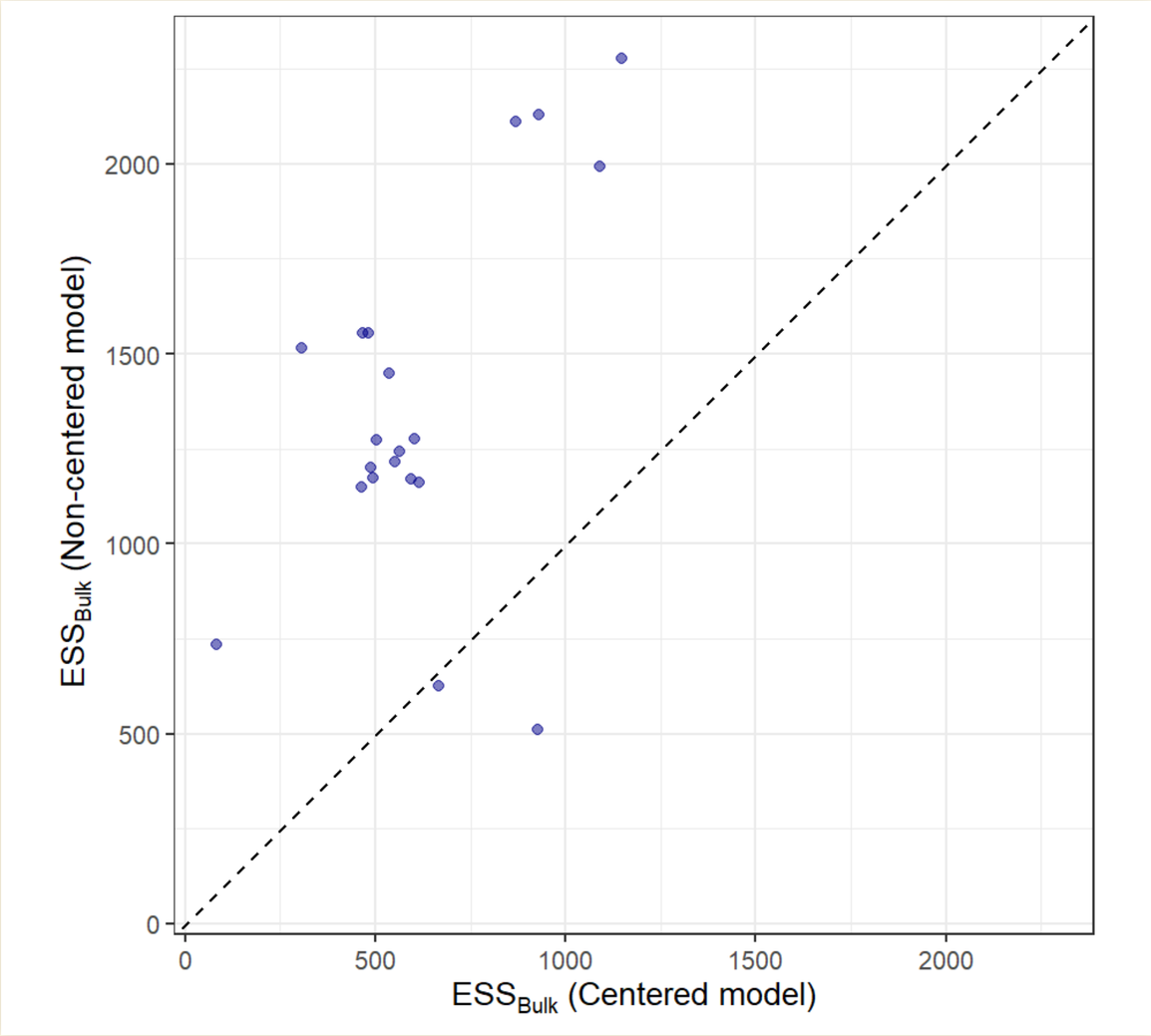
Coding the non-centered model

```
set.seed(13)
mdl_chimp_nc <- ulam(
    alist(
        pulled_left ~ dbinom(1, p) ,
        logit(p) <- a_bar + z[actor] * sigma_a +
                    x[block_id] * sigma_g +
                    b[treatment],
        b[treatment] ~ dnorm(0, 0.5),
        z[actor] ~ dnorm(0, 1),
        x[block_id] ~ dnorm(0, 1),
        a_bar ~ dnorm(0, 1.5),
        sigma_a ~ dexp(1),
        sigma_g ~ dexp(1),
        gq> vector[actor]:a <<- a_bar + z * sigma_a,
        gq> vector[block_id]:g <<- x * sigma_g
    ), data = dat_list, chains = 4, cores = 4)
precis(mdl_chimp_nc, digits = 2)
```

```
##         mean   sd  5.5% 94.5% rhat ess_bulk
## a_bar   0.62 0.74 -0.51  1.79 1.01   509.90
## sigma_a 2.06 0.67  1.22  3.28 1.00   625.83
## sigma_g 0.20 0.16  0.02  0.50 1.01   734.24
```

```
mdl_chimp_nc@cstanfit$diagnostic_summary("divergences")
```

```
## $num_divergent
## [1] 0 0 0 0
```



Comparing Results

```
precis(mdl_chimp.4, depth = 2, digits = 2)
```

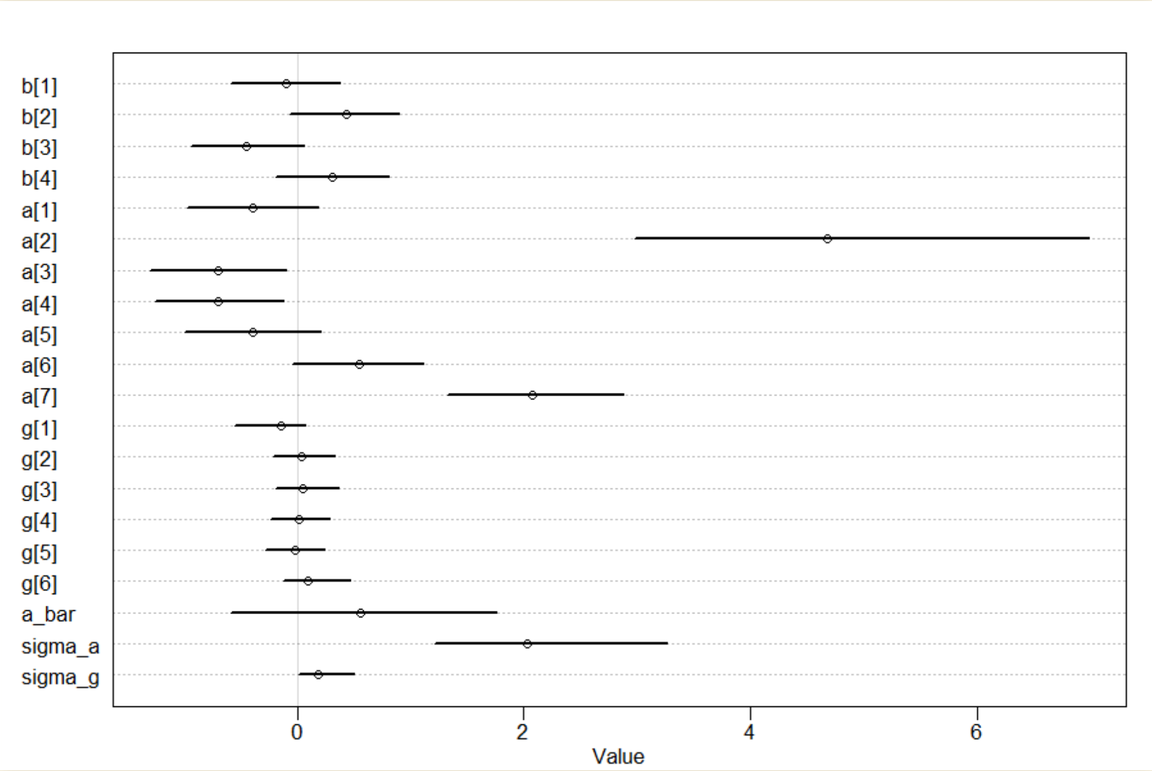
```
##          mean   sd  5.5% 94.5% rhat ess_bulk
## b[1]    -0.11 0.31 -0.59  0.38 1.02   338.77
## b[2]     0.43 0.30 -0.07  0.90 1.01   343.49
## b[3]    -0.45 0.31 -0.94  0.06 1.01   442.27
## b[4]     0.31 0.31 -0.19  0.81 1.01   447.73
## a[1]    -0.39 0.36 -0.97  0.18 1.01   518.25
## a[2]     4.68 1.43  2.98  6.99 1.01   648.44
## a[3]    -0.70 0.38 -1.30 -0.10 1.01   435.22
## a[4]    -0.70 0.37 -1.26 -0.13 1.01   460.22
## a[5]    -0.39 0.37 -0.99  0.20 1.01   439.61
## a[6]     0.54 0.37 -0.04  1.11 1.01   437.74
## a[7]     2.08 0.48  1.33  2.87 1.01   542.35
## g[1]    -0.14 0.22 -0.56  0.07 1.01   405.53
## g[2]     0.03 0.18 -0.22  0.33 1.02   809.69
## g[3]     0.05 0.18 -0.19  0.36 1.02   646.92
## g[4]     0.01 0.17 -0.23  0.28 1.02   831.97
## g[5]    -0.02 0.18 -0.29  0.24 1.01   886.57
## g[6]     0.10 0.19 -0.12  0.47 1.01   636.02
## a_bar    0.56 0.74 -0.59  1.76 1.00   896.99
## sigma_a  2.03 0.69  1.22  3.26 1.00   851.11
## sigma_g  0.18 0.18  0.01  0.50 1.03   107.82
```

```
precis(mdl_chimp_nc, depth = 2, digits = 2)
```

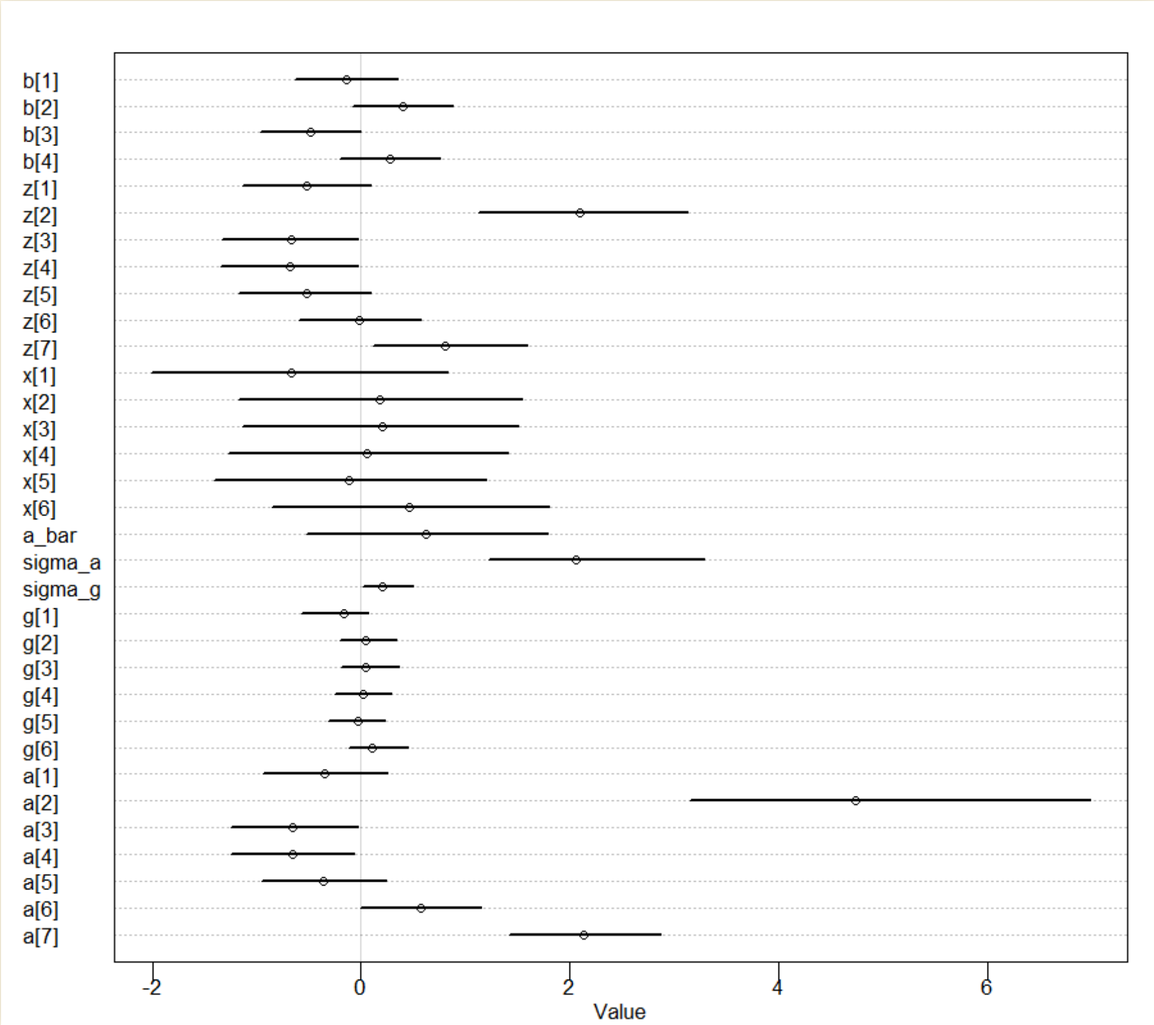
```
##          mean   sd  5.5% 94.5% rhat ess_bulk
## b[1]    -0.14 0.31 -0.63  0.35 1.00  1162.45
## b[2]     0.40 0.30 -0.08  0.88 1.00  1171.39
## b[3]    -0.48 0.30 -0.97 -0.01 1.00  1244.50
## b[4]     0.27 0.30 -0.20  0.76 1.00  1200.03
## z[1]    -0.51 0.39 -1.14  0.10 1.01   460.24
## z[2]     2.09 0.63  1.13  3.13 1.00  1010.57
## z[3]    -0.67 0.41 -1.33 -0.03 1.01   463.71
## z[4]    -0.67 0.41 -1.34 -0.03 1.01   435.74
## z[5]    -0.52 0.39 -1.18  0.10 1.01   452.48
## z[6]    -0.02 0.37 -0.60  0.57 1.01   497.16
## z[7]     0.81 0.46  0.12  1.59 1.00   622.35
## x[1]    -0.67 0.89 -2.01  0.83 1.00  2253.53
## x[2]     0.19 0.87 -1.17  1.55 1.00  2006.40
## x[3]     0.21 0.83 -1.13  1.51 1.00  2486.69
## x[4]     0.06 0.83 -1.26  1.40 1.00  1870.83
## x[5]    -0.11 0.84 -1.41  1.21 1.00  2073.95
## x[6]     0.46 0.85 -0.86  1.80 1.00  1946.22
## a_bar    0.62 0.74 -0.51  1.79 1.01   509.90
## sigma_a  2.06 0.67  1.22  3.28 1.00   625.83
## sigma_g  0.20 0.16  0.02  0.50 1.01   734.24
## g[1]    -0.16 0.21 -0.57  0.07 1.00  1516.18
## g[2]     0.04 0.18 -0.20  0.33 1.00  2112.49
## g[3]     0.05 0.18 -0.19  0.36 1.00  2278.83
## g[4]     0.02 0.17 -0.25  0.29 1.01  1993.91
## g[5]    -0.03 0.17 -0.31  0.23 1.01  2128.82
## g[6]     0.11 0.19 -0.11  0.45 1.00  1554.64
## a[1]    -0.35 0.37 -0.94  0.26 1.00  1271.86
## a[2]     4.73 1.32  3.15  6.99 1.00  1555.23
## a[3]    -0.65 0.38 -1.25 -0.03 1.00  1275.61
## a[4]    -0.66 0.37 -1.25 -0.07 1.00  1174.44
## a[5]    -0.36 0.37 -0.95  0.24 1.00  1150.48
## a[6]     0.58 0.36 -0.01  1.15 1.00  1215.21
## a[7]     2.13 0.46  1.42  2.87 1.00  1449.84
```

Comparing Results

```
plot(precis(mdl_chimp.4, depth = 2, digits = 2))
```




```
plot(precis(mdl_chimp_nc, depth = 2, digits = 2))
```



Back to the Radon Example

Multilevel Radon Model

```
ml_data <- select(mn_radon, county_id, log_radon, floor)


mdl_ml_1 <- ulam(
  alist(
    log_radon ~ dnorm(mu, sigma),
    mu <- a[county_id] + b * floor,
    a[county_id] ~ dnorm(a_bar, sig_a),
    b ~ dnorm(0, 10),
    sigma ~ dexp(1),
    a_bar ~ dnorm(0, 10),
    sig_a ~ dexp(1)
  ), data = ml_data, chains = 4, cores = 4, log_lik = TRUE)

precis(mdl_ml_1, digits = 2, depth = 2,
       pars = c("a_bar", "sig_a", "b", "sigma",
                paste0("a[", 1:5, "]")))
```

```
##        mean   sd  5.5% 94.5% rhat ess_bulk
## a_bar  1.46 0.05  1.38  1.54 1.00   987.00
## sig_a  0.33 0.05  0.26  0.41 1.00   768.82
## b     -0.69 0.07 -0.80 -0.59 1.00  1393.98
## sigma  0.76 0.02  0.73  0.78 1.00  2419.80
## a[1]   1.19 0.25  0.79  1.58 1.01  2851.87
## a[2]   0.93 0.10  0.77  1.09 1.00  2181.96
## a[3]   1.48 0.27  1.04  1.91 1.00  2730.30
## a[4]   1.51 0.22  1.15  1.86 1.00  2473.64
## a[5]   1.45 0.26  1.05  1.86 1.01  2887.43
```

- Variation in α represents effects of soil uranium and home construction.

```
ml_data_2 <- select(mn_radon, county_id, log_radon, floor,
        log_uranium)

mdl_ml_2 <- ulam(
  alist(
    log_radon ~ dnorm(mu, sigma),
    mu <- a[county_id] + b * floor + g * log_uranium,
    a[county_id] ~ dnorm(a_bar, sig_a),
    b ~ dnorm(0, 2),
    sigma ~ dexp(1),
    a_bar ~ dnorm(0, 2),
    sig_a ~ dexp(1),
    g ~ dnorm(0, 2)
  ), data = ml_data_2, chains = 4, cores = 4, log_lik = TRUE)

precis(mdl_ml_2, digits = 2, depth = 2,
       pars = c("a_bar", "sig_a", "b", "sigma", "g",
                paste0("a[", 1:5, "]")))
```

```
##        mean   sd  5.5% 94.5% rhat ess_bulk
## a_bar  1.47 0.04  1.40  1.53 1.00   512.25
## sig_a  0.16 0.05  0.08  0.24 1.01   116.30
## b     -0.67 0.07 -0.78 -0.56 1.00  1370.74
## sigma  0.76 0.02  0.73  0.79 1.00  1839.65
## g      0.72 0.09  0.58  0.86 1.00  1638.50
## a[1]   1.45 0.15  1.20  1.67 1.00  2226.16
## a[2]   1.48 0.10  1.31  1.63 1.00  1911.11
## a[3]   1.48 0.16  1.23  1.73 1.01  2607.32
## a[4]   1.58 0.16  1.34  1.84 1.00   576.29
## a[5]   1.48 0.15  1.25  1.72 1.00  2312.81
```

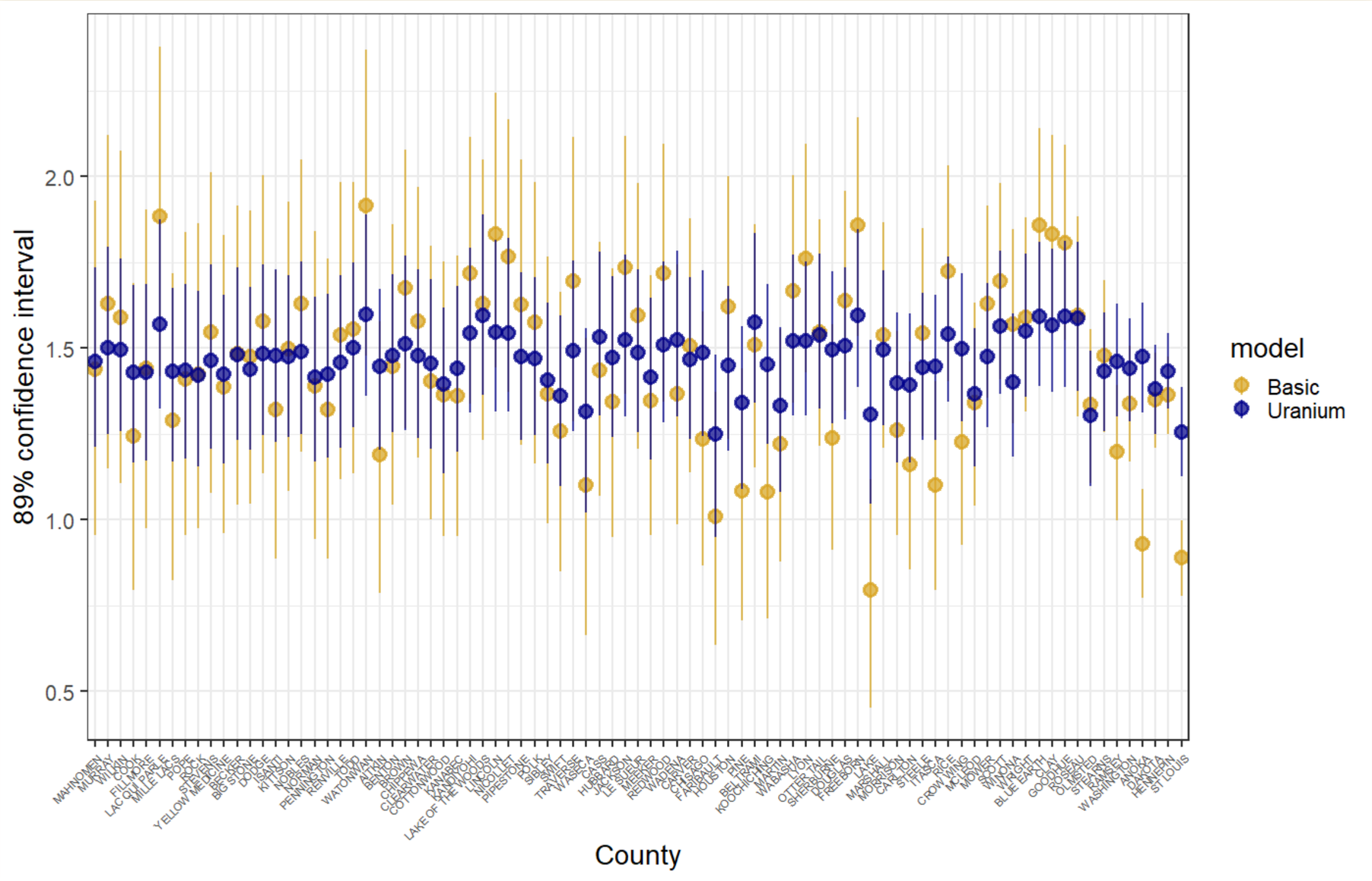
- Variation in α only represents effects of home construction.

Compare Models

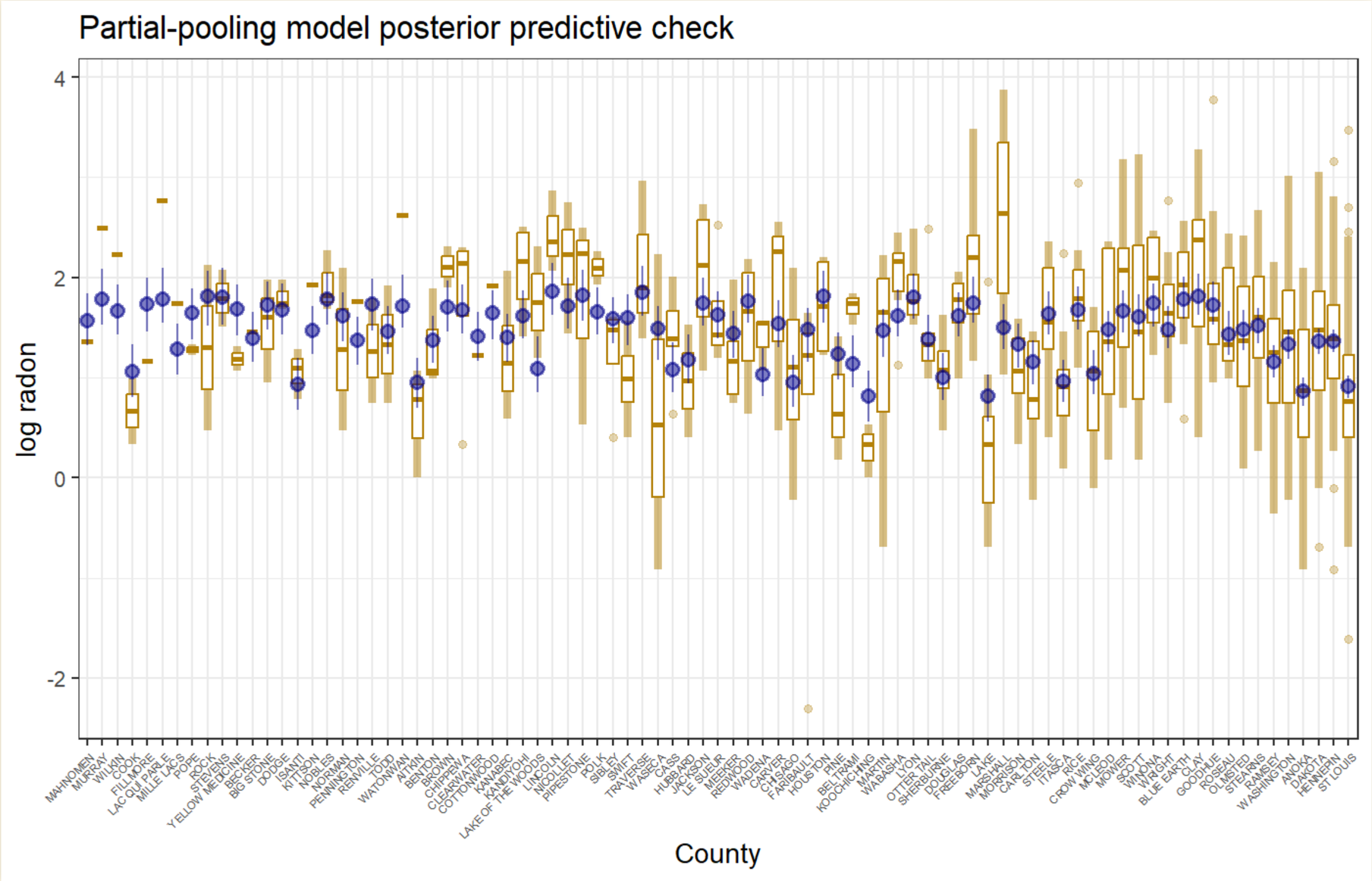
```
ml_1_draws <- gather_draws(mdl_ml_1, a[county_id]) |> ungroup() |>
  left_join(select(mn_counties, county_id, county), by = "county_id") |>
  mutate(model = "Basic")

ml_1_sum <- ml_1_draws |> group_by(county, county_id, model) |>
  summarize(mean = mean(.value), PI = list(set_names(PI(.value), c("lower", "upper")))) |>
  unnest_wider(PI)

ml_2_draws <- gather_draws(mdl_ml_2, a[county_id]) |> ungroup() |>
  left_join(select(mn_counties, county_id, county, log_uranium),
            by = "county_id") |>
  mutate(model = "Uranium")

ml_2_sum <- ml_2_draws |> group_by(county, county_id, model) |>
  summarize(mean = mean(.value), PI = list(set_names(PI(.value), c("lower", "upper")))) |>
  unnest_wider(PI)
```

```
ggplot(ml_2_sum, aes(x = county, y = mean, ymin = lower, ymax = upper, color = model)) +
  geom_pointrange(data = ml_1_sum, alpha = 0.7) +
  geom_pointrange(alpha = 0.7) +
  scale_color_manual(values = c(Uranium = "darkblue", Basic = "goldenrod")) +
  labs(x = "County", y = "89% confidence interval", title = ) +
  theme(axis.text.x = element_text(size = 8, angle = 45, hjust = 1, vjust = 1))
```

Compare Models



Posterior Prediction Check (Uranium Model)



Post-Stratification

Post-Stratification

- Statistical analysis when we know our data come from a non-representative sample
  - We know N_i, the relative populations of different groups i
    - But our data is not distributed proportionately to N_i
  - **Examples:**
    - Public-opinion surveys contact people
      - Some demographic groups are more likely than others to answer surveys
    - Crowd-sourcing biodiversity studies:
      - Crowd-sourced wildlife censuses produce large data, but over- or under-count different species
      - Expert censuses accurately measure populations, but only for small areas.
- **MRP** (Mr. P): **M**ultilevel **R**egression modeling with **P**oststratification
  - Analyze non-representative sample using multilevel regression:
    - Partial pooling helps estimate effects for under-counted groups:
      - μ_i is the effect for group i
  - Apply *poststratification* to the results:
    - Predict representative population-level effects $\mu_{\text{population}}$ by weighting regression results μ_i for different groups by relative abundance of those groups in the whole population.

$$\mu_{\text{population}} = \frac{\sum_i N_i \mu_i}{\sum_i N_i}$$